Aggregated Pyramid Vision Transformer: Split-transform-merge Strategy for Image Recognition without Convolutions

Rui-Yang Ju[1], Ting-Yu Lin[2], Jen-Shiun Chiang[1], Jia-Hao Jian[1], Yu-Shian Lin[1], and Liu-Rui-Yi Huang[1]
Tamkang University, New Taipei City, Taiwan[1]
National Cheng Kung University, Tainan, Taiwan[2]

2022 IEEE International Conference on Consumer Electronics - Taiwan | 978-1-6654-7050-6/22/$31.00 ©2022 IEEE | DOI: 10.1109/ICCE-TAIWAN55306.2022.9869242

***Abstract*-- This paper proposes a new pure attention model, Aggregated Pyramid Vision Transformer (APVT), for computer vision applications. Based on the Vision Transformer (ViT) architecture, APVT adopts the classic pyramid architecture of CNN and employs the group encoder technique to replace the traditional encoder for feature enhancement. APVT uses the split-transform-merge strategy to refine the group encoder operation. The model performs image classification on CIFAR-10 dataset and object detection on COCO 2017 dataset for verification. Experimental results show that APVT has excellent performance compared to other Transformer network architectures.**

I. Introduction

Vision Transformer (ViT) [1] is the first technology to apply the entire Transformer to computer vision (CV) tasks. Although the experimental results do not achieve state-of-the-art (SOTA), ViT inspired researchers to use Transformer as the primary block in network architecture. Pyramid Vision Transformer (PVT) [2] realizes that ViT suffers from the disadvantage in high computational cost and low resolution of output feature maps, and therefore in turn uses a pyramid model to overcome these difficulties.

However, despite PVT solves the problem of high computational cost, it still has the disadvantage of the fact that the Transformer architecture fails to fully extract image features. Therefore, this research work proposes a new pure attention model, Aggregated Pyramid Vision Transformer (APVT), which uses the split-transform-merge strategy in ResNeXt [3] and proposes the mechanism of group encoder to replace the traditional encoder to obtain image features more efficiently. To reduce the model size, we set all paths in each aggregate Transformer to share the same topology.

The remainder of this paper is organized as follows. We describe the system framework in details in Section 2, give experimental results in Section 3, and draw conclusions in Section 4.

II. Aggregated Pyramid Vision Transformer

A. Architecture

APVT is a pure Transformer framework that introduces a pyramid architecture, which consists of multiple improved Transformers as the primary block. Its overall architecture is shown in Fig. 1. We divide the APVT network architecture into four stages and obtain features of different sizes through the four stages. Each aggregated Transformer (stage) consists of the same architecture, including patch embedding and group encoder, which are connected in a similar way of residual learning architecture, with position embedding added in the architecture. For an input image of size Height (H) × Width (W), the first patch embedding would obtain features of $H/4 \times W/4$; the number of patches is $H \times W/16$, and then the second patch embedding would obtain $1/2 \times 1/2$ features on this basis. For the original input image, the features are $H/8 \times W/8$. The rest of the features can be obtained in the same manner.

B. Split-transform-merge

In order to achieve a low error rate and low computational cost, APVT applies the split-transform-merge strategy to the Transformer, where *split* is to divide the vector into low-level embedding representations, *transform* is to undergo a linear transformation for each low-dimensional feature, and *merge* to be the final output through unit addition.

C. Group Encoder

According to the split-transform-merge strategy, we can divide the traditional encoder into multiple identical (group) encoders, and then perform operations on each group encoder to operate more efficient feature extraction on the input image, and finally fuse all the group encoders. This work attempts to split the encoder of ViT into multiple group encoders. The splitting and fusion methods are shown in Fig. 2. The aggregated Transformer is composed of group encoders as the primary block, which not only extracts more features, but also improves the accuracy, without adding too much size to the network model.

D. Model Details

As shown in Table 1, we set the Heads to Emb values to 32 and 64 respectively, corresponding to the number of heads to determine the embedding size. For example, the embedding sizes of APVT-8-2x-a are 32, 64, 160 and 256, respectively. In addition, we set n paths to 2 and 3, respectively. Different models may use 2 or 3 group encoders.

III. Experiment Results

For experiment, we employ RTX 3080 Ti to test image classification for APVT and other Transformer network architectures. A small APVT-8-2x-a model is used for object detection tasks.

A. Image Classification

Image classification experiments are performed on the CIFAR-10 dataset, and all models are not pre-trained for fair comparisons. In training, the batch size is set to 128. The initial learning rate is set to 5×10^{-4}, and is reduced by 10% every 30 epochs. We use AdamW [5] to optimize the model. All models are run for 60 epochs on one single RTX 3080Ti.

Table 2 shows that APVT model outperforms other Transformer network architectures under similar numbers of parameters, and APVT-8-2x-b has the lowest error rate of 20.39% among models with smaller numbers of parameters. Compared with PVTv2-B1, the inference time of APVT-8-2x-b

Authorized licensed use limited to: Fondren Library Rice University. Downloaded on October 06,2023 at 14:40:50 UTC from IEEE Xplore. Restrictions apply.

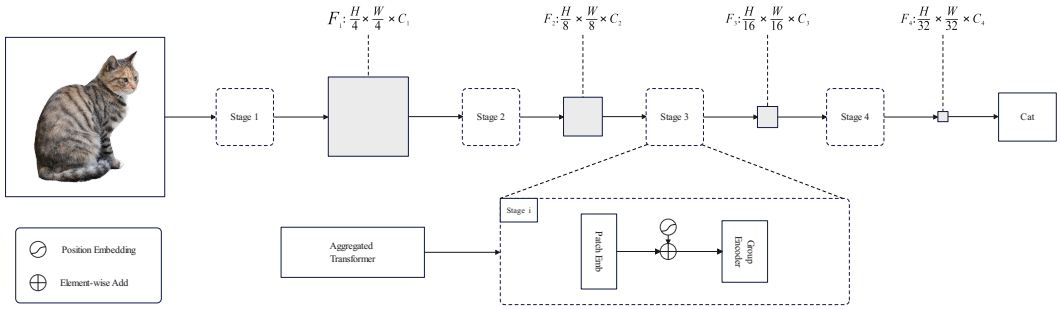

Fig. 1. The overall architecture of Aggregated Pyramid Vision Transformer.

is reduced from 25ms to 17ms. In the comparison of models with larger numbers of parameter, APVT-16-2x achieves the best results, with an error rate of 19.55%.

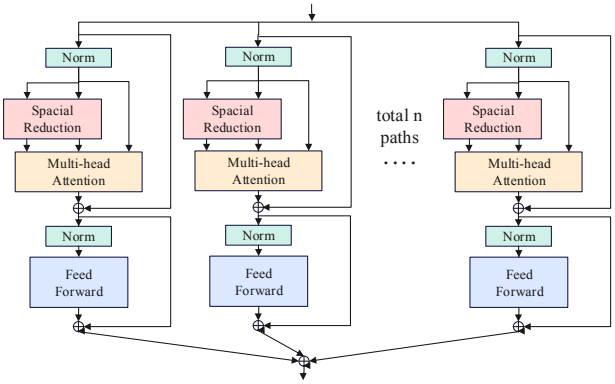

Fig. 2. The architecture of the group encoder. Total *n* paths are *n* group encoders, and *n* are set to 2 and 3, respectively.

Model	Layers	Paths	Head to Emb	Heads	Params
APVT-8-2x-a	2, 2, 2, 2	2	32	1, 2, 5, 8	5.52M
APVT-8-2x-b	2, 2, 2, 2	2	64	1, 2, 5, 8	22.88M
APVT-16-2x-b	3, 4, 6, 3	2	64	1, 2, 5, 8	42.15M
APVT-8-4x-a	2, 2, 2, 2	3	32	1, 2, 5, 8	8.16M
APVT-16-4x-a	3, 4, 6, 3	3	32	1, 2, 5, 8	15.48M

Table 1. Details of Aggregated Pyramid Vision Transformer.

Name	#Param (M)	CIFAR-10 Err (%)	Inference Time (ms)
LeViT-256	17.89	29.48	20
BoTNet-50	18.81	23.03	14
CvT-13	19.54	21.23	21
CeiT-S	21.17	31.28	19
APVT-8-2x-b	22.88	**20.39**	**17**
DeepViT-S	23.66	39.71	27
PVTv2-B1	25.41	21.22	25
BoTNet-101	32.22	22.35	25
LeViT-384	37.63	28.03	23
APVT-16-2x-b	42.15	**19.55**	**28**
DeepViT-B	50.45	36.56	33

Table 2. Image classification performance on CIFAR-10. Our network results are in **boldface**, surpassing all competing methods in blue.

B. *Object Detection*

Object detection experiments are performed on the COCO 2017 dataset, where the model is not pre-trained, and we use the standard detector RetinaNet [6] to test the performance of APVT-8-2x-a. We train with batch size of 16 and optimize by AdamW [5] with an initial learning rate set to 1×10^{-4}. Our model is trained for 12 epochs on one single RTX 3080Ti.

As shown in Table 3, we recorded the average precision (AP) values at 6 epochs and 12 epochs, respectively, which are 10.4 and 16.8, respectively. For a small model APVT with a parameter size of 5.52M, an AP of 16.8 is obtained without pre-training indicateing our excellent performance.

Method	Epoch	AP	AP_{50}	AP_{75}	AP_S	AP_M	AP_L
RetinaNet	6	10.4	18.0	10.6	5.3	11.4	13.5
	12	16.8	27.2	17.6	8.7	17.9	21.7

Table 3. Object detection performance on COCO val2017. The model is trained for 6/12 epochs using the RetinaNet method, and the backbone is APVT-8-2x-a. The results of object detection are not pre-trained in advance.

IV. Conclusion

As Transformer has achieved excellent results in the field of CV, many convolutional neural networks add Transformer to their networks to improve model efficiency, and there are also many neural networks that abandon the convolutional layer and use Transformer as the primary block. Experimental results show that our proposed aggregated Transformer outperforms the traditional Transformer, which not only achieves good results in image classification, but also performs outstandingly in object detection tasks.

Reference

[1] A. Dosovitskiy *et al.*, "An image is worth 16x16 words: Transformers for image recognition at scale," *International Conference on Learning Representations*, 2021.

[2] W. Wang *et al.*, "Pyramid Vision Transformer: A versatile backbone for dense prediction without convolutions," *International Conference on Computer Vision*, Oct. 2021, pp. 568-578.

[3] S. Xie, R. Girshick, P. Dollar, Z. Tu, and K. He, "Aggregated residual transformations for deep neural networks," *IEEE Conference on Computer Vision and Pattern Recognition*, July. 2017, pp. 1492-1500.

[4] W. Wang *et al.*, "PVTv2: Improved baselines with pyramid vision transformer," *arXiv:2106.13797*, 2022.

[5] I. Loshchilov and F. Hutter, "Decoupled weight decay regularization," International *Conference on Learning Representations*, 2019.

[6] T. Lin, P. Goyal, R. Girshick, K. He, and P. Dollar, "Focal loss for dense object detection" *IEEE Transactions on Pattern Analysis & Machine Intelligence*, vol. 42, no. 02, pp. 318-327, 2020.

Authorized licensed use limited to: Fondren Library Rice University. Downloaded on October 06,2023 at 14:40:50 UTC from IEEE Xplore. Restrictions apply.